Introduction to the Program Design and Programming

ICT 1305

CHAPTER 03

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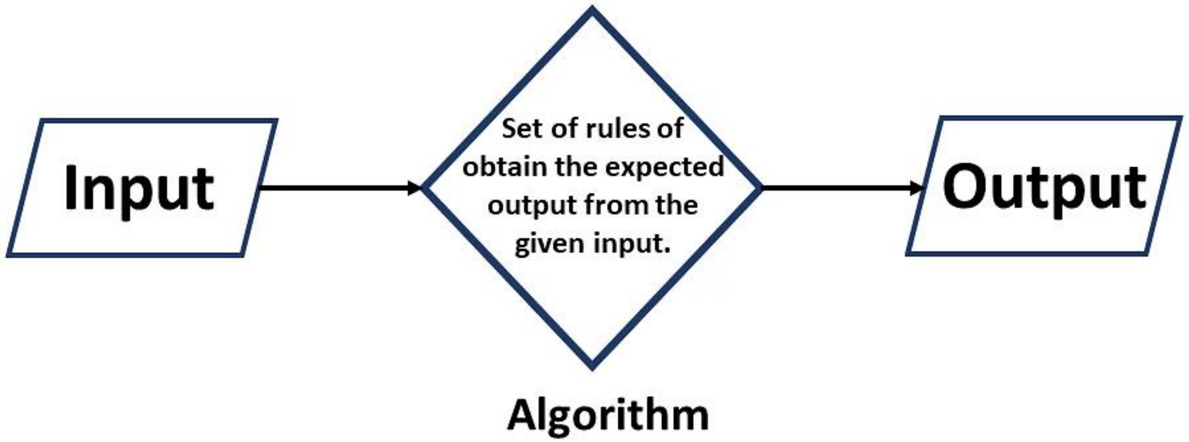
What is an Algorithm

- Algorithms are the threads that tie together most of the subfields of computer science.
- An algorithm is a step-by-step procedure or set of rules designed to perform a specific task or solve a particular problem.
- It is a logical and systematic way of accomplishing an objective, often used in mathematics, computer science, and other fields.



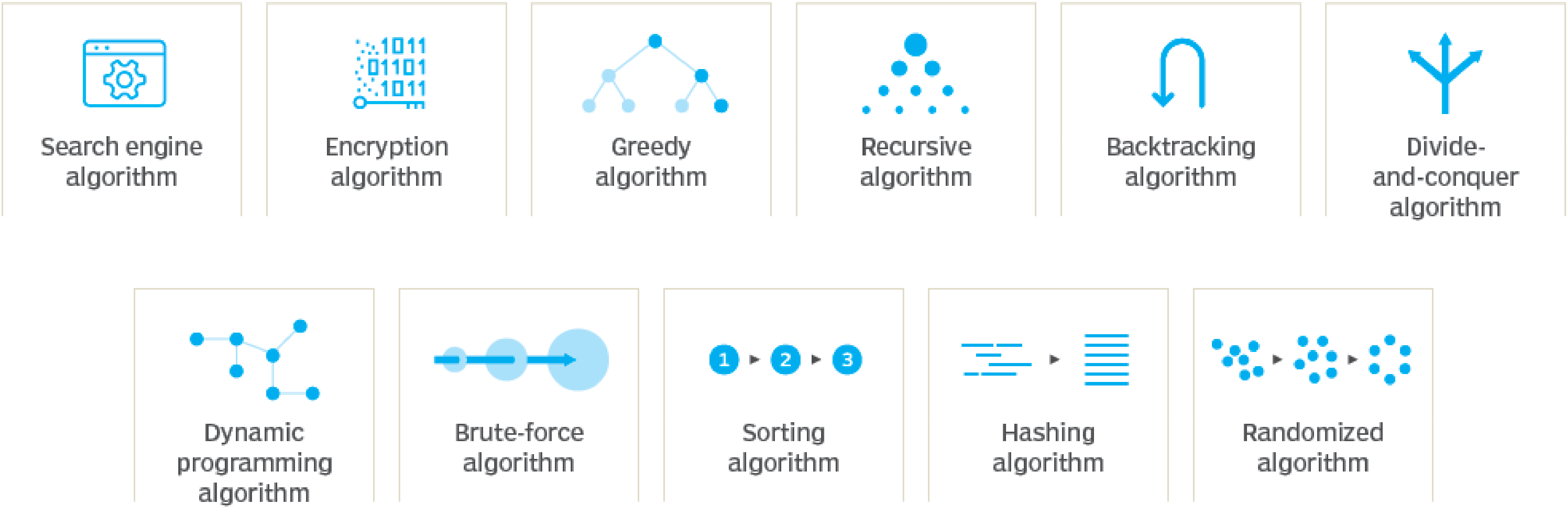
How do Algorithms Work?

- **Input:** Algorithms take input data, which can be in various formats, such as numbers, text, or images.
- **Processing:** The algorithm processes the input data through a series of logical and mathematical operations, manipulating and transforming it as needed.
- **Output:** After the processing is complete, the algorithm produces an output, which could be a result, a decision, or some other meaningful information.
- **Efficiency:** A key aspect of algorithms is their efficiency, aiming to accomplish tasks quickly and with minimal resources.
- **Optimization:** Algorithm designers constantly seek ways to optimize their algorithms, making them faster and more reliable.
- **Implementation:** Algorithms are implemented in various programming languages, enabling computers to execute them and produce desired outcomes.

Properties of an Algorithm

- **Finiteness:** The algorithm must always terminate after a finite number of steps.
- **Definiteness:** Each step must be precisely defined; the actions to be carried out must be rigorously and unambiguously specified for each case.
- **Input:** An algorithm has zero or more inputs, taken from a specified set of objects.
- **Output:** An algorithm has one or more outputs, which have a specified relation to the inputs.
- **Effectiveness:** All operations to be performed must be sufficiently basic that they can be done exactly and in finite length.

Types of Algorithms



Types of Algorithms

- **Brute Force Algorithm:** A straightforward approach that exhaustively tries all possible solutions, suitable for small problem instances but may become impractical for larger ones due to its high time complexity.
- **Recursive Algorithm:** A method that breaks a problem into smaller, similar subproblems and repeatedly applies itself to solve them until reaching a base case, making it effective for tasks with recursive structures.
- **Encryption Algorithm:** Utilized to transform data into a secure, unreadable form using cryptographic techniques, ensuring confidentiality and privacy in digital communications and transactions.

Types of Algorithms

- **Searching Algorithm:** Designed to find a specific target within a dataset, enabling efficient retrieval of information from sorted or unsorted collections.
- **Greedy Algorithm:** Makes locally optimal choices at each step in the hope of finding a global optimum, useful for optimization problems but may not always lead to the best solution.
- **Divide and Conquer Algorithm:** Breaks a complex problem into smaller subproblems, solves them independently, and then combines their solutions to address the original problem effectively.
- **Dynamic Programming Algorithm:** Stores and reuses intermediate results to avoid redundant computations, enhancing the efficiency of solving complex problems.

Types of Algorithms

- **Randomized Algorithm:** Utilizes randomness in its steps to achieve a solution, often used in situations where an approximate or probabilistic answer suffices.
- **Backtracking Algorithm:** A trial-and-error technique used to explore potential solutions by undoing choices when they lead to an incorrect outcome, commonly employed in puzzles and optimization problems.
- **Hashing Algorithm:** Converts data into a fixed-size hash value, enabling rapid data access and retrieval in hash tables, commonly used in databases and password storage.
- **Sorting Algorithm:** Aimed at arranging elements in a specific order, like numerical or alphabetical, to enhance data organization and retrieval.

Use of the Algorithms

- **Data Analysis and Machine Learning**
  - Algorithms in data analysis and machine learning identify patterns in large datasets and predict outcomes.
  - Techniques like support vector machines, decision trees, and neural networks enable computers to learn and improve, supporting applications such as recommendation systems, natural language processing, and image recognition.
- **Cryptography & Security**
  - Cryptography algorithms, like SHA-256, AES, and RSA, protect data through encryption and decryption, ensuring secure storage, communication, and online transactions.
  - They underpin cyber security measures for data integrity, user authentication, and sensitive information security.

Use of the Algorithms

- **Information Retrieval and Search Engines**
  - Search algorithms like PageRank and Hummingbird enable search engines to index and retrieve relevant information efficiently.
  - They rank web pages by importance and relevance, helping users find the most pertinent content amid vast online data.
- **Optimization problems**
  - Optimization methods, such as gradient descent, linear programming, and genetic algorithms, identify the best solutions to complex problems.
  - Widely used in industries like finance, engineering, logistics, and AI, they enhance productivity, reduce costs, and optimize resource use for operations and decision-making.

Use of the Algorithms

- **Genomics and medical diagnostics**

  - Algorithms play a vital role in medical diagnostics by analyzing images, predicting disease outbreaks, and identifying genetic changes.
  - Machine learning has revolutionized personalized medicine, enabling tailored treatments based on genetic profiles, while also accelerating genomic data sequencing and interpretation, advancing biotechnology and genomics research.

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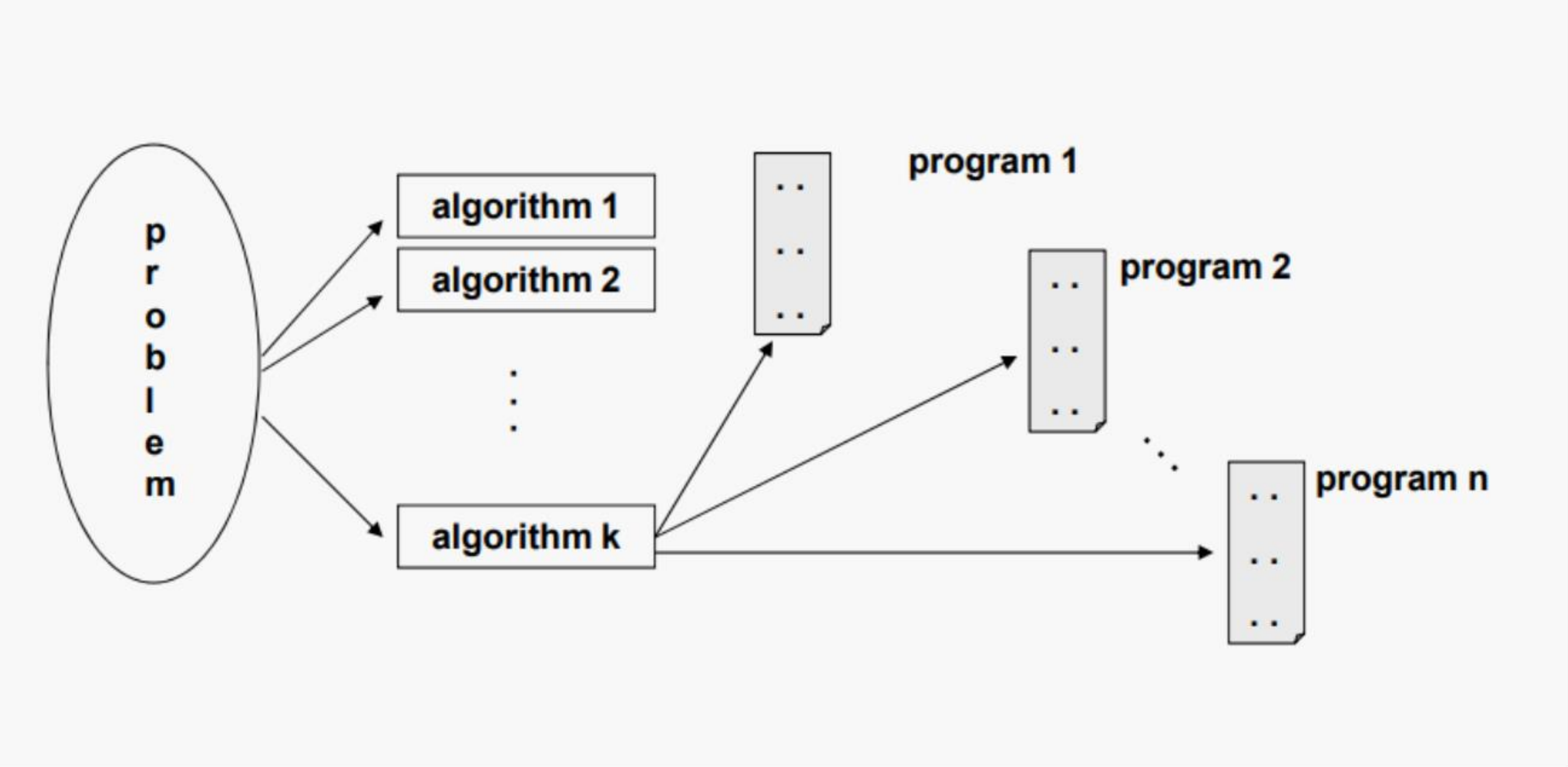
Problems vs Algorithms vs Programs

- For each problem or class of problems, there may be many different algorithms.
- For each algorithm, there may be many different implementations (programs).
- A concrete implementation of an algorithm in a programming language that can be executed by a computer.
- The machine can’t read a program directly, because it only understands machine code.
- Write stuff in a computer language, and then a compiler or interpreter can make it understandable to the computer.

Problems vs Algorithms vs Programs



Expressing Algorithms

- **Natural Language:** usually verbose and ambiguous
- **Flowcharts:** avoid most (if not all) issues of ambiguity; difficult to modify without specialized tools; largely standardized
- **Pseudo-code:** also avoids most issues of ambiguity; vaguely resembles common elements of programming languages; no particular agreement on syntax
- **Programming Language:** tend to require expressing low-level details that are not necessary for a high-level understanding

Common Elements of Algorithms

Acquire data (input)

some means of reading values from an external source; most algorithms require data values to define the specific problem

Computation

some means of performing arithmetic computations, comparisons, testing logical conditions, and so forth.

Selection

some means of choosing among two or more possible courses of action, based upon initial data, user input and/or computed results

Common Elements of Algorithms

Iteration

some means of repeatedly executing a collection of instructions, for a fixed number of times or until some logical condition holds

Report results (output)

some means of reporting computed results to the user, or requesting additional data from the user

How to Write an Algorithm?

- There are no well-defined standards for writing algorithms.
- An algorithm can be written using these common basic code constructs (do, for, while, if, else).
- Algorithms are typically written in a step-by-step fashion.
- Algorithm writing is a process that occurs after the problem domain has been well-defined.
    1) Understand the Problem
    2) Plan the Logic
    3) Write the Steps
    4) Verify and Test the algorithm

Algorithm to make a cup of tea

start boil the water , tea packet, if need add mil, add sugger , mix end

START-----BOIL WATER------ADD TEA BAG------POUR WATER IN CUP-------REMOVE TEA BAG-------ADD SUGER AND MILK---------STIR--------DRINK---------STOP

add some water to the mouth , after that add some milk powder as well , after add some sugar and jump 10 times continuously , tea is ready

1.start 2.Taking a cup and tea bag 3. boiling water 4. put the boiled water in a cup 5. remove the tea bag 6. Then add sugar 7. stir in a cup of tea.

Algorithm to make a Cup of Tea

1. Fill up the kettle with water
2. Boil the water
3. Place a teabag in your favorite mug
4. Pour boiling water into your favorite mug
5. Brew the tea for a few moments
6. Remove and dispose of the teabag
7. Add milk
8. Add sugar
9. Stir the tea
10. Enjoy the hot beverage

Algorithm to send an Email

start login gmail , click compose, writ receiver email , write subject if need, write massage , if need attached files send ,close the gmail account, end

go to the web site, click the mail button, type the mail name we want to send, and type something, and send it.

start-go to the email app-enter the subject-enter the receiver address- write the compose-send-stop

Start - open Gmail app - click compose email - add email address - type subject - compose email - attach file -send email - Stop

1.on the data connection 2. go to our mail account 3.Then open to compose email 5. entry is the sender's email 6.entry is the subject details 7.then entry is a body details 8. add to the send document or file 9.finally send to the mail.

start login gmail , click compose, writ receiver email , write subject if need, write massage , if need attached files send ,close the gmail account, end

go to the web site, click the mail button, type the mail name we want to send, and type something, and send it.

Algorithm to send an email

1. Open an internet browser and go to the email provider's website.
2. Click on the "Sign in" button and enter your email address and password when prompted.
3. Click on the "Compose" or "New email" button to open a new email message.
4. In the "To" field, enter the email address of the person you want to send the email to.
5. In the "Subject" field, write a brief description of what the email is about.
6. In the main body of the email, write your message.
7. Add an attachment (If needed).
8. Click on the "Send" button.
9. Check your sent folder to make sure that the email was sent successfully.

Let’s Apply Algorithmic concepts into Computing

Build an Algorithm

Algorithm for Add 10 and 20

1. To solve this problem we will take a variable sum and set it to zero.
2. Then we will take the two numbers 10 and 20 as input.
3. Next we will add both the numbers and save the result in the variable sum i.e., sum = 10 + 20.
4. Finally, we will print the value stored in the variable sum.

Build an Algorithm

Algorithm (in simple English)

1. Initialize sum = 0 (PROCESS)
2. Enter the numbers (I/O)
3. Add them and store the result in sum (PROCESS)
4. Print sum (I/O)

Build an Algorithm

Algorithm for Find the sum of 5 numbers

1. Take two variables - sum and count and set both of them to zero.
2. The sum variable will store the result while the count variable will keep track of how many numbers have read.
3. To solve this problem, use the concept of loop.
4. In loop or iterative operation, execute some steps repeatedly as long as the given condition is TRUE.
5. Initialize sum and count to zero. Then we will take the input and store it in a variable **n**.
6. Next add the value stored in n to sum and save the answer in sum.
7. Increase count by 1 and check if count is less than 5. If this condition is TRUE then take another input. If the condition is FALSE then print the value stored in variable sum.

Build an Algorithm

1. Initialize sum = 0 and count = 0 (PROCESS)
2. Enter n (I/O)
3. Find sum + n and assign it to sum and then increment count by 1 (PROCESS)
4. Is count < 5 (DECISION)
5. if YES go to step 2
6. Else
7. Print sum (I/O)

Build an Algorithm

Algorithm for Print Hello World 10 times

1. This problem is also solved using the loop concept.
2. Take a variable count and set it to zero.
3. Then print "Hello World" and increment count by 1.
4. Next, check if count is less than 10. If this is TRUE then again print "Hello World" and increment the variable count.
5. On the other hand if the condition if FALSE then execution will stop.

Build an Algorithm

1. Initialize count = 0 (PROCESS)
2. Print Hello World (I/O)
3. Increment count by 1 (PROCESS)
4. Is count < 10 (DECISION)
5. if YES go to step 2
6. else Stop

Build an Algorithm

Algorithm to find the average of 3 subjects

Step 1: Start the Program

Step 2: Declare and Read 3 Subject, (S1, S2, S3)

Step 3: Calculate the sum of all the 3 Subject values and store result in Sum variable

(Sum = S1+S2+S3)

Step 4: Divide Sum by 3 and assign it to Average variable. (Average = Sum/3)

Step 5: Print the value of Average of 3 Subjects

Step 6: End the Program

Build an Algorithm

Algorithm for multiplies two numbers and displays the output.

Step 1 – Start

Step 2 – declare three integers x, y & z

Step 3 – define values of x & y

Step 4 – multiply values of x & y

Step 5 – store result of step 4 to z

Step 6 – print z

Step 7 – Stop

Thank You